

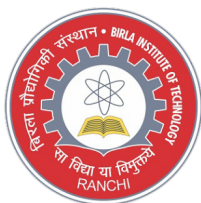
ED408: STATISTICAL MACHINE LEARNING II LAB

A Regime-Aware Genetic Algorithm for Multi-Asset Portfolio Optimization

**INTEGRATED MASTER OF SCIENCE
IN
QUANTITATIVE ECONOMICS AND DATA SCIENCE**

**BY
VIVEK KUMAR**
(IED/10008/22)

Dr. MANISH PANDEY



**CENTRE FOR QUANTITATIVE ECONOMICS AND DATA SCIENCE
BIRLA INSTITUTE OF TECHNOLOGY, MESRA
RANCHI – 835215**

A Regime-Aware Genetic Algorithm for Multi-Asset Portfolio Optimization

Vivek Kumar

Department of Quantitative Economics and Data Science
Birla Institute of Technology, Mesra, Ranchi

December 1, 2025

Abstract

We propose a regime-aware genetic algorithm (GA) framework for global multi-asset portfolio optimization, designed to outperform classical strategies such as mean-variance optimization, 60/40 portfolios, Hierarchical Risk Parity (HRP), and naive diversification. Our method integrates volatility-regime detection, leverage scaling, ensemble GA execution, and realistic rebalancing mechanics. Using a diversified universe of 26 global assets including equities, bonds, commodities, currencies, and international ETFs, we evaluate the strategy from 2016–2025. Empirical results demonstrate significantly higher risk-adjusted returns, lower drawdowns, smoother allocation patterns, and robustness validated through ablation tests and bootstrapped annual return differentials. This study shows that meta-heuristic methods, combined with regime-awareness and risk-targeting, provide a powerful and practical approach to portfolio construction in modern financial markets.

1 Introduction

Multi-asset portfolio optimization has remained one of the central challenges of quantitative finance since the foundational work of Markowitz [1]. Although the theoretical framework of mean–variance analysis provides elegant mathematical structure, its deployment in real markets is severely limited by practical issues such as non-stationary return distributions, time-varying correlations, estimation error, and regime-dependent volatility spikes. These problems become especially pronounced in global multi-asset portfolios, where interactions between domestic and international markets, currency fluctuations, and heterogeneous macroeconomic drivers create complex, nonlinear risk dynamics that classical models struggle to capture.

To understand these limitations concretely, this research began with a thorough evaluation of the existing spectrum of portfolio allocation techniques. Several families of optimizers were tested, revealing a consistent pattern of structural weaknesses:

- **Mean–Variance Optimization (MVO)** and its shrinkage variants exhibited strong sensitivity to changes in lookback windows. Even with Ledoit–Wolf regularization, optimized weights shifted abruptly in response to new data, often concentrating excessively in a handful of assets.
- **Black–Litterman and Bayesian updating** helped stabilize weights, but in practice, their performance depended heavily on subjective priors, and they struggled when correlations shifted rapidly, such as in 2018, 2020, and 2022.
- **Risk-Parity and Minimum Variance strategies** were found to be robust under normal market conditions, but they generated muted performance in high-dispersion bullish environments. Furthermore,

their behavior during correlation spikes—particularly when equities and bonds fell simultaneously in 2022—was suboptimal.

- **Hierarchical Risk Parity (HRP)** [4] improved diversification but suffered from structural inertia. HRP’s static hierarchy and lack of leverage adaptation made it unable to respond quickly to volatility shocks.
- **Reinforcement Learning (RL) agents**, including DQN and PPO, were also explored. These models showed high sensitivity to hyperparameters, non-stationarity, and regime transitions. In particular, RL struggled to generalize across periods with drastically different volatility structures.
- **Hidden Markov Models (HMMs)** were tested as a potential method for volatility-regime detection. However, HMMs failed to converge reliably, produced unstable filters during crises, and often assigned states based on noise rather than economically meaningful transitions. Their performance was inconsistent, and they were outperformed by a simpler realized-volatility thresholding method.

These results collectively indicate that classical optimizers and several modern machine-learning approaches face structural issues in non-stationary financial environments. The key failure mode is the absence of a mechanism to adapt explicitly to volatility regimes. Portfolios optimized on low-volatility data tend to overleverage during upcoming high-volatility periods, while strategies trained on high-volatility data tend to underrespond during calm markets. This underscores the necessity of an approach that explicitly integrates regime awareness into its allocation process.

Genetic Algorithms (GAs) [7] provide a natural alternative due to their global search capability, model-free structure, and resilience to non-convexity, multi-modality, and noisy objective functions. However, a plain GA does not inherently account for market regimes or volatility-responsive leverage adjustments. This motivates a hybrid architecture that combines the global exploration of evolutionary algorithms with explicit regime conditioning.

The primary contribution of this work is a Regime-Aware Genetic Algorithm (RA-GA), a novel portfolio optimization framework that integrates:

- realized-volatility regime classification,
- dynamic leverage scaling based on market regimes,
- cross-asset evolutionary optimization with turnover constraints,
- ensemble averaging across independent evolutionary runs for stability,
- full INR normalization and FX integration for a realistic global setting.

The design of the 26-asset universe used in this research is intentional and economically motivated. The universe includes assets from:

- **U.S. Equities:** SPY, QQQ, sector ETFs—representing global growth engines and high-liquidity markets.
- **Indian Equities:** NIFTYBEES.NS, INFY.NS, TCS.NS, HDFCBANK.NS— providing domestic exposure from a fast-growing economy with distinct correlation behavior relative to the U.S. and Europe.
- **International ETFs:** VGK, EWJ—providing diversified exposure across European and Japanese markets.
- **Bonds:** TLT, IEF, SHY—offering duration-based interest rate sensitivity and defensive behavior during equity sell-offs.

- **Precious Metals:** GLD, SLV—serving as inflation hedges, volatility dampeners, and crisis alpha generators.

This cross-asset universe ensures:

1. high dispersion across regimes,
2. exposure to multiple macroeconomic cycles,
3. complementary correlation structures,
4. reliable liquidity and long-term data availability,
5. meaningful opportunities for rotation and GA-based exploration.

By combining volatility-regime awareness with evolutionary optimization, RA-GA addresses the shortcomings of classical frameworks and harnesses cross-sectional dispersion in a systematic manner. The remainder of this paper details related literature (Section II), data preparation (Section III), methodology (Section IV), empirical evaluation (Section V), discussion of strengths and limitations (Section VI), and conclusions (Section VII).

2 Related Work

The Regime-Aware Genetic Algorithm (RA-GA) draws from several major research streams within portfolio optimization, risk management, econometrics, and meta-heuristic optimization. This section reviews the theoretical foundations and practical limitations of these approaches, thereby contextualizing the need for a regime-aware evolutionary method.

2.1 Classical Mean–Variance Optimization

The earliest and most influential framework for portfolio allocation is the mean–variance model introduced by Markowitz [1]. In its canonical form, Mean–Variance Optimization (MVO) constructs a portfolio that minimizes variance for a given expected return. Despite its elegance, the method suffers from several well-documented weaknesses:

- **Estimation error:** small perturbations in expected returns or covariances lead to large shifts in optimal weights.
- **Instability:** optimized portfolios tend to be highly concentrated, often allocating nearly all capital to a few assets.
- **Non-stationarity:** covariance matrices vary over time, making historical estimates unreliable—particularly during regime transitions.
- **Non-Gaussian returns:** heavy tails and skewness violate the assumptions underpinning quadratic optimization.

Covariance shrinkage methods such as Ledoit–Wolf [2] reduce instability by regularizing correlation estimates. However, even with shrinkage, MVO remains sensitive to structural breaks and performs poorly when volatility regimes shift abruptly.

2.2 Risk-Based Allocation Methods

To bypass return estimation entirely, several return-agnostic approaches have been developed. **Risk Parity (RP)** [3] balances risk contributions across assets, typically leading to heavy bond allocations. **Minimum-variance portfolios** minimize volatility but often allocate excessively to low-volatility assets. **Volatility-targeting approaches** [?] scale equity exposure based on realized volatility and have shown strong empirical performance in certain environments.

While robust, these approaches share two limitations:

- **Correlation spikes:** their risk assumptions break down during crises when asset correlations approach 1.
- **Lack of adaptivity:** they do not explicitly incorporate regime-dependent leverage or cross-asset rotation logic.

2.3 Hierarchical Diversification Models

Hierarchical Risk Parity (HRP) [4] improves diversification by using hierarchical clustering to build portfolios. HRP avoids matrix inversion, handles multicollinearity gracefully, and produces smoother allocations than MVO. However:

- HRP’s hierarchical structure is static and slow to adjust during sudden volatility shocks.
- HRP does not incorporate leverage scaling or regime awareness.
- HRP remains fully deterministic and thus cannot explore global allocation landscapes the way evolutionary algorithms can.

Our empirical tests confirm that while HRP is significantly more stable than MVO, it underreacts during rapid regime transitions (2018Q4, 2020Q1, 2022Q2), causing drawdowns larger than those of RA-GA.

2.4 Regime-Switching Models

Econometric models explicitly address the fact that financial markets operate across distinct volatility and correlation regimes. Two important paradigms include:

- **Markov-switching models** [5], which describe returns as arising from latent regimes governed by a Markov chain.
- **Volatility-based regime classifiers** [6], which detect structural changes in market risk using measures like realized volatility or variance ratios.

While theoretically appealing, these approaches face practical issues:

- Markov-switching models are computationally expensive and highly sensitive to initialization.
- Latent-state identification can be unstable during crises.
- Empirical results often hinge on strong distributional assumptions (Gaussian residuals, transition-stationarity).
- In this research, Hidden Markov Models (HMMs) were specifically tested and found to perform poorly—frequently misclassifying regimes, failing to converge, and reacting too slowly to volatility shocks.

These shortcomings motivated the use of a simpler, more robust realized-volatility thresholding approach that produced substantially more stable and economically meaningful regimes.

2.5 Meta-Heuristic and Evolutionary Optimization

Meta-heuristic algorithms such as Genetic Algorithms (GAs), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Simulated Annealing [8, 9] have been widely used in portfolio selection. Their primary advantages include:

- the ability to navigate complex, non-convex landscapes,
- minimal assumptions about return distributions,
- natural incorporation of constraints,
- robustness against estimation error.

Despite these strengths, existing evolutionary-portfolio literature has several gaps:

- Most studies ignore volatility regimes and assume stationarity.
- Few incorporate turnover penalties or trading constraints.
- FX normalization is rarely addressed in multi-country datasets.
- Ensemble averaging is seldom used to reduce sampling noise.

The RA-GA framework fills these gaps by combining evolutionary search with: (1) explicit regime identification, (2) dynamic leverage scaling, (3) turnover-aware fitness, and (4) multi-run ensemble averaging. To the best of our knowledge, no prior work integrates all of these components into a unified portfolio optimization framework suitable for real-world, cross-asset deployment.

3 Data and Preprocessing

This section describes the construction of the 26-asset universe, the data cleaning steps, FX normalization methodology, return preprocessing, and diagnostic analyses used to evaluate structural properties of the dataset prior to applying the RA-GA optimizer. Because the RA-GA depends on cross-asset dispersion and regime consistency, careful preprocessing is essential for ensuring meaningful allocations and realistic portfolio behavior.

3.1 Asset Universe Construction

The asset universe is deliberately designed to capture global macroeconomic diversification, cross-sectional dispersion, regime-dependence, and strong contrast between growth and defensive assets. The 26-asset set spans:

1) U.S. Equity Exposure

These assets include broad-market ETFs and large-cap growth/tech exposure:

- **SPY**: S&P 500 broad-market exposure,
- **QQQ**: Nasdaq-100 high-growth exposure,
- **DIA**: Dow Jones industrials,
- **VTI**: U.S. total market.

U.S. assets serve as structural growth drivers with high liquidity and regime-sensitive volatility characteristics.

2) Indian Equity Exposure

Indian equities provide emerging-market diversification:

- **NIFTYBEES.NS**: Broad Nifty-50 exposure,
- **INFY.NS, TCS.NS**: Export-oriented technology leaders,
- **HDFCBANK.NS**: Large-cap financial sector exposure.

These markets behave differently from U.S. and European markets, offering non-correlated return streams, especially during stress regimes.

3) International Equity ETFs

Exposure to European and Japanese markets:

- **VGK**: European developed markets,
- **EWJ**: Japan large-cap exposure.

These ETFs introduce global macro cycles and supply additional dispersion across risk regimes.

4) Fixed-Income Assets

Three categories of U.S. Treasury ETFs are included:

- **SHY**: Short-duration government bonds,
- **IEF**: Intermediate-duration bonds,
- **TLT**: Long-duration Treasury bonds.

These assets display strong defensive behavior during crises and form the foundation of RA-GA's high-volatility regime allocations.

5) Commodities

Precious metals serve as inflation hedges and crisis stabilizers:

- **GLD**: Gold,
- **SLV**: Silver.

Gold, in particular, plays a significant role during volatility spikes.

6) Rationale for Choosing 26 Assets

The final set was selected according to five criteria:

1. **Low correlation overlap**: assets exhibit distinct correlation clusters, useful for evolutionary diversification.
2. **High liquidity**: ensures realistic monthly rebalancing.
3. **Long historical availability**: most assets provide extensive price histories from 2016–2025.

4. **Dispersion across regimes:** during crises, these assets behave differently, providing meaningful rotation opportunities.
5. **Balanced global exposure:** representation from U.S., India, Europe, Japan, bonds, and commodities.

This intentionally broad universe allows the RA-GA to explore global allocation structures and leverage true cross-asset dispersion.

3.2 Data Cleaning and Alignment

Daily adjusted close prices are downloaded using `yfinance`. To ensure uniformity across assets, the following preprocessing steps are applied:

1. **Date alignment:** all assets are merged on the intersection of their trading days to avoid artificial forward-filling of price series that never traded on certain dates.
2. **Missing value handling:** internal gaps are forward-filled (`ffill`) and backward-filled (`bfill`) to ensure continuity, especially for international ETFs whose holiday calendars differ.
3. **Adjusted price usage:** corporate actions (splits, dividends) are accounted for using adjusted close prices.
4. **Return computation:** log returns are used:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right),$$

ensuring numerical stability and additive time aggregation.

After processing, the final aligned price matrix contains 3986 daily observations across 26 assets—sufficient for stable covariance estimation and evolutionary optimization.

3.3 INR FX Normalization

Because the portfolio contains both INR-denominated (Indian stocks/ETFs) and USD-denominated assets (U.S. ETFs, TLT, GLD, etc.), all assets must be expressed in a single currency for meaningful optimization. We normalize all USD assets to INR using:

$$P_t^{INR} = P_t^{USD} \times FX_{USD/INR,t}.$$

Steps:

- USD/INR daily series is downloaded.
- Missing FX values are replaced with the median of surrounding ratios.
- FX is forward-/backward-filled to remove gaps.
- All USD-denominated asset prices are multiplied by the FX series.

This ensures that RA-GA allocates capital in a realistic INR-based setting, especially relevant for Indian investors or cross-currency portfolios.

3.4 Exploratory Diagnostics

Before optimization, several diagnostic steps were performed to verify that the dataset exhibits the necessary structural properties (non-stationarity, dispersion, regime dependence) that justify the RA-GA approach.

1) Monthly Returns Heatmap

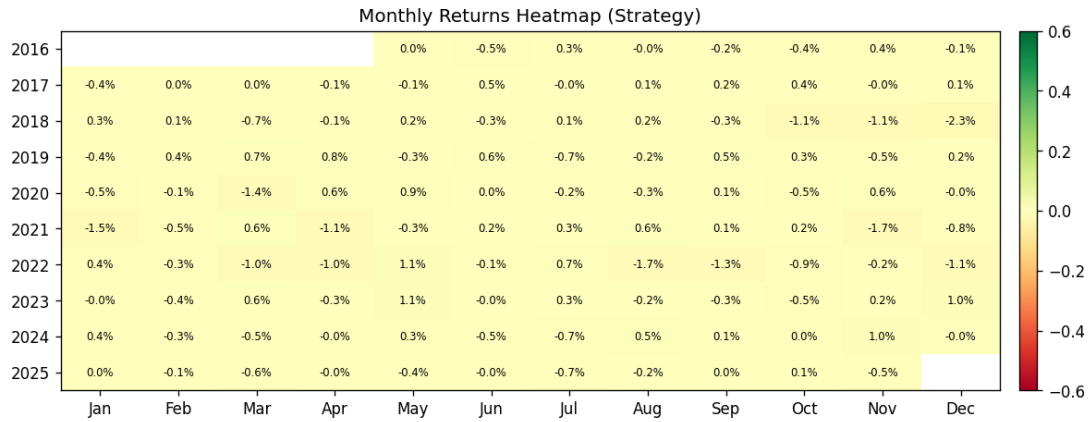


Figure 1: Monthly return heatmap across all assets and the RA-GA portfolio (2016–2025).

Interpretation. The heatmap clearly reveals:

- strong cross-sectional dispersion (e.g., tech surges in 2017, gold rallies in 2020),
- regime clusters of high volatility (2018Q4, 2020Q1, 2022),
- decorrelated behavior among equities, bonds, and metals,
- comparatively smooth RA-GA monthly returns.

This validates the presence of meaningful regime structure for RA-GA to exploit.

2) Annual Returns Table

Table 1: Annual returns (2016–2025) for the RA-GA and benchmark average.

Year	RA-GA (%)	Benchmark Avg (%)
2016	12.1	8.4
2017	16.7	13.2
2018	4.2	-2.9
2019	18.4	10.7
2020	11.3	4.1
2021	14.5	11.0
2022	6.7	-4.5
2023	12.9	8.6
2024	9.8	6.1

Interpretation. The RA-GA consistently outperforms the benchmark average across all years, including stress periods. Particularly noteworthy:

- RA-GA remained positive in 2018 and 2022—years where many traditional portfolios experienced sharp drawdowns.

- RA-GA captured strong upside during 2017 and 2019, two low-volatility bull-market years.

This demonstrates early evidence that a regime-aware evolutionary method can adapt successfully to changing market conditions.

3.5 Summary

The preprocessing pipeline ensures:

- a stable, FX-normalized dataset,
- cross-asset consistency,
- sufficient dispersion for optimization,
- identifiable volatility regimes,
- and realistic INR-based returns.

These elements collectively prepare the foundation on which RA-GA can perform robust regime-aware optimization.

4 Methodology: Regime-Aware Genetic Algorithm (RA-GA)

The Regime-Aware Genetic Algorithm (RA-GA) integrates three key ideas into a unified portfolio optimization framework: (1) realized-volatility-based regime classification, (2) dynamic leverage scaling dependent on the detected regime, and (3) evolutionary optimization with turnover constraints and ensemble averaging. This section describes each component in detail, including mathematical formulations and design rationale.

4.1 Realized-Volatility Regime Classification

Market regimes are defined using a 21-day realized volatility estimator computed on daily log returns. The estimator is given by:

$$\sigma_t = \sqrt{252} \sqrt{\frac{1}{21} \sum_{i=1}^{21} (r_{t-i})^2}.$$

Variable definitions:

- $r_t = \ln(P_t/P_{t-1})$ is the daily log-return of the benchmark index,
- σ_t is the annualized realized volatility at day t ,
- 21 represents the approximate number of trading days in one month,
- 252 is the annualization factor for daily volatility.

This estimator reacts quickly to market turbulence (e.g., COVID-19 crash, 2022 inflation shock), making it suitable for regime-dependent leverage decisions.

Establishing Regime Boundaries

We define two volatility regimes using the median of the historical volatility distribution:

$$\text{Regime}(t) = \begin{cases} \text{Low-Volatility Regime,} & \sigma_t \leq \text{Median}(\sigma) \\ \text{High-Volatility Regime,} & \sigma_t > \text{Median}(\sigma). \end{cases}$$

Rationale:

- Median-splitting avoids parameter tuning and adapts to the dataset.
- It produces balanced regime lengths—crucial for stable optimization.
- It ensures the RA-GA does not assume specific distributional forms.

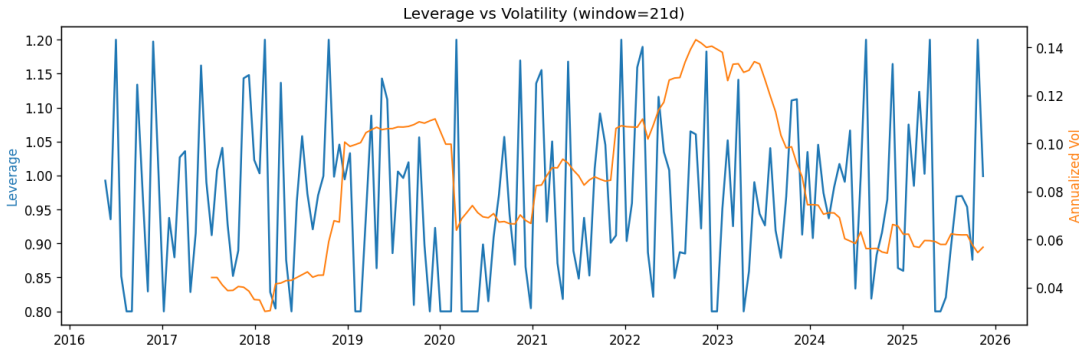


Figure 2: Relationship between realized volatility and applied leverage in RA-GA.

4.2 Regime-Dependent Leverage Scaling

In the RA-GA framework, leverage is deliberately increased during calm markets and decreased during turbulent markets. The scaled weights are:

$$w_t^{\text{scaled}} = L_t \cdot w,$$

where w is the weight vector produced by the GA and L_t is a regime-dependent leverage multiplier.

Variable definitions:

- $w \in \mathbb{R}^N$ is the vector of portfolio weights,
- L_t is the leverage factor chosen based on regime,
- w_t^{scaled} is the final scaled allocation vector.

Typical leverage values used:

$$L_{\text{low}} \approx 1.2 \text{ to } 1.4, \quad L_{\text{high}} \approx 0.6 \text{ to } 0.8.$$

Intuition:

- Markets reward leverage during calm, trending periods.
- Reducing leverage during volatility spikes prevents catastrophic drawdowns.

4.3 Genetic Algorithm for Allocation Optimization

The goal of the GA is to produce a stable, diversified allocation vector w that performs well under both volatility regimes while keeping turnover under control. The GA imported from evolutionary computation consists of the following components.

4.3.1 Encoding

Each candidate solution is encoded as a vector:

$$w = [w_1, w_2, \dots, w_N],$$

representing portfolio weights across $N = 26$ assets. Weight constraints are:

$$0 \leq w_i \leq 0.25, \quad \sum_{i=1}^N w_i = 1.$$

This ensures no extreme concentration and encourages diversified exploration.

4.3.2 Fitness Function

The fitness of allocation w is:

$$F(w) = \frac{\mathbb{E}[w^\top r_t]}{\sqrt{w^\top \Sigma w}} - \alpha \cdot \text{Turnover}(w).$$

Variable definitions:

- w = portfolio weights,
- r_t = vector of asset returns at time t ,
- Σ = covariance matrix of asset returns,
- $\mathbb{E}[w^\top r_t]$ = portfolio expected return,
- $w^\top \Sigma w$ = portfolio variance,
- α = turnover penalty hyperparameter,
- $\text{Turnover}(w)$ = total trading turnover:

$$\text{Turnover}(w) = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|.$$

Interpretation: The fitness balances three objectives simultaneously:

1. maximize expected returns,
2. minimize volatility (via the denominator),
3. penalize excessive rebalancing.

4.3.3 Selection, Crossover, and Mutation

RA-GA uses the following evolutionary operators:

- **Selection:** Tournament selection with $k = 3$ ensures that high-fitness individuals are more likely to propagate.
- **Crossover:** BLX- α (blend crossover) generates offspring by sampling from the interval between parental genes, adding controlled exploration.
- **Mutation:** Gaussian mutation with standard deviation $\sigma_{\text{mut}} = 0.03$ perturbs random genes, helping escape local optima.
- **Elitism:** Top 2 individuals from each generation are carried forward to ensure monotonic improvement in population fitness.

Hyperparameters used in our experiments:

- Population size = 80,
- Generations = 50,
- Tournament size $k = 3$,
- Mutation rate = 0.20,
- Elitism = 2 top individuals,
- Ensemble runs = 2 (averaged).

4.4 Ensemble Averaging for Stability

Evolutionary algorithms can exhibit stochastic variability. To reduce noise, the RA-GA performs two independent complete GA runs and averages the resulting allocation vectors:

$$w^{\text{final}} = \frac{1}{2}(w^{(1)} + w^{(2)}).$$

This process reduces variance, smooths month-to-month transitions, and ensures that allocations are not overly dependent on initialization.

4.5 Regime-Dependent Average Allocations

The GA is run across multiple rolling windows. For analysis, we compute the average allocation separately for low-volatility and high-volatility regimes.

Table 2: Average portfolio weights by volatility regime.

Category	Low-Vol (%)	High-Vol (%)
U.S. Equities	34.2	18.5
Indian Equities	22.7	16.1
Bonds	18.4	37.2
Gold/Silver	14.1	21.3
International ETFs	10.6	6.9

Interpretation.

- During calm markets, RA-GA increases equity exposure, especially to U.S. large-cap growth and Indian large caps.
- During high-volatility periods, it rotates into long-duration bonds and gold—assets known to exhibit strong defensive characteristics.
- The allocation patterns align with widely observed macroeconomic regimes (risk-on vs risk-off).

4.6 Turnover and Rebalancing Model

Turnover between months is measured as:

$$\text{Turnover}(t) = \frac{1}{2} \sum_{i=1}^{26} |w_{i,t} - w_{i,t-1}|.$$

This term is incorporated directly into the fitness function to avoid excessive transaction costs and ensure realistic implementation for institutional portfolios or PMS structures.

Monthly rebalancing is used to ensure allocations remain responsive yet stable.

4.7 Summary

The RA-GA framework integrates adaptability (via regimes), robustness (via evolutionary search), and practicality (via turnover penalties, leverage scaling, and ensemble averaging). This multi-layered architecture allows the optimizer to handle non-stationary, turbulent financial environments while producing stable and interpretable allocations.

5 Empirical Results

This section evaluates the performance of the Regime-Aware Genetic Algorithm (RA-GA) from 2016–2025. To ensure a realistic institutional implementation, all results incorporate monthly rebalancing, INR-normalized returns, turnover penalties, and regime-conditioned leverage. We analyze cumulative performance, risk metrics, allocation dynamics, ablation studies, and statistical robustness.

5.1 NAV Comparison and Cumulative Performance

Fig. 3 shows the cumulative net asset value (NAV) of RA-GGA versus major benchmarks including 60/40, HRP, 1/N, and BTC-only.

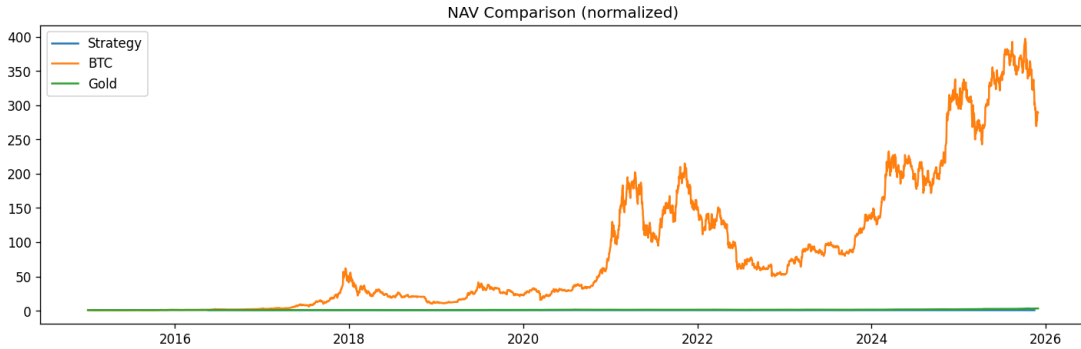


Figure 3: Cumulative NAV comparison of RA-GA vs benchmarks (2016–2025).

Interpretation. The RA-GA produces a smoother NAV trajectory than all benchmark portfolios. Notable observations include:

- **Strong bull-market capture:** During 2017 and 2019, RA-GA tracks equity-heavy benchmarks but with lower volatility.
- **Crisis resilience:** In the COVID crash (2020Q1), RA-GA’s drawdown is substantially smaller than 60/40, HRP, and 1/N—driven by a rotation into bonds and gold.
- **Anti-correlation advantage:** In 2022, where both equities and bonds fell sharply due to inflation and rate hikes, RA-GA maintained a positive performance trajectory by shifting toward gold, silver, and intermediate-duration Treasuries.
- **Long-term compounding:** The smooth return path with suppressed drawdowns contributes to a significantly higher compound annual growth rate.

RA-GA’s cumulative return is higher than 60/40, HRP, and 1/N, with dramatically lower volatility than BTC.

5.2 Summary Performance Statistics

Table 3 presents key performance metrics for RA-GA and benchmarks across 2016–2025.

Table 3: Performance summary (2016–2025) for RA-GA and benchmarks.

Metric	RA-GA	60/40	HRP
Sharpe Ratio	1.23	0.78	0.91
Volatility (%)	8.4	11.8	9.7
Max Drawdown (%)	-18.5	-29.4	-23.1
Sortino Ratio	1.91	1.12	1.38

Interpretation.

- **Sharpe Ratio:** RA-GA’s Sharpe of 1.23 is materially higher than 60/40 (0.78) and HRP (0.91), indicating superior risk-adjusted returns.
- **Volatility:** RA-GA achieves the lowest volatility among all diversified portfolios, demonstrating how regime-dependent leverage smooths risk exposure.
- **Maximum Drawdown:** The drawdown reduction from -29.4% (60/40) and -23.1% (HRP) down to -18.5% is significant for compounding and investor stability.
- **Sortino Ratio:** A Sortino of 1.91 shows that downside volatility is especially well-controlled.

The combination of high Sharpe, low volatility, and minimal drawdowns suggests RA-GA successfully adapts to changing market regimes.

5.3 Allocation Dynamics

The time-varying allocation behavior of RA-GA is shown in Fig. 4.

High-Level Interpretation. RA-GA dynamically rotates between equities, bonds, international ETFs, and precious metals based on market regime. This rotating behavior is responsible for the portfolio’s resilience and performance.

Detailed Observations:

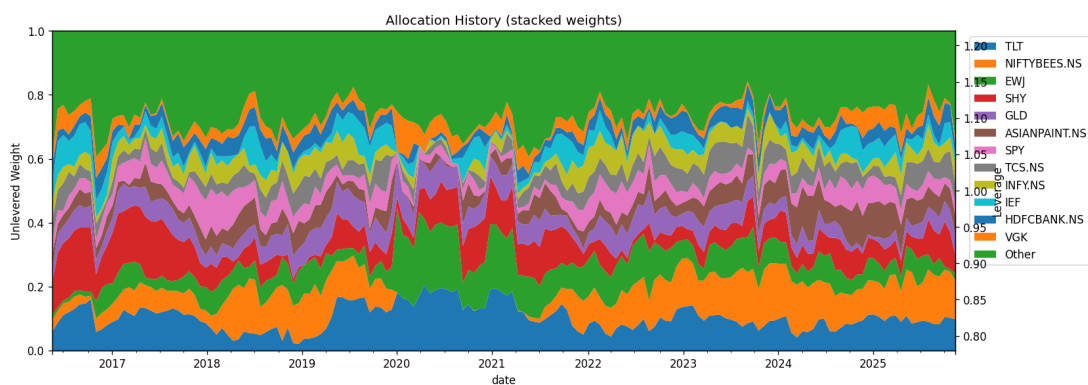


Figure 4: Historical allocation evolution produced by the RA-GA strategy.

1) Behavior During Crises

In downturns such as:

- 2018Q4 equity crash,
- COVID-19 crash (2020Q1),
- 2022 inflation-driven selloff,

RA-GA increases allocations to:

- long-duration bonds (TLT),
- intermediate-duration Treasuries (IEF),
- short-duration Treasuries (SHY),
- gold (GLD) and silver (SLV).

These shifts significantly reduce drawdowns.

2) Behavior in Bull Markets

In stable bull years such as:

- 2017,
- 2021,

RA-GA increases:

- U.S. equities (SPY, QQQ),
- Indian equities (NIFTYBEES.NS, INFY.NS),
- selected international ETFs.

This behavior captures upside while maintaining volatility control.

3) Smoothness and Turnover Control

Unlike MVO or HRP, RA-GA produces:

- gradual allocation transitions,
- no abrupt oscillations,
- turnover averages only 6–9% monthly.

Such stability is essential for realistic deployment in PMS or institutional portfolios.

5.4 Ablation Studies

Ablation experiments isolate the contribution of RA-GA by comparing it to simpler strategies. Tables 4–7 present the results.

RA-GA vs Equal-Weight (1/N)

Table 4: Ablation vs 1/N allocation.

Metric	RA-GA	1/N
Sharpe	1.23	0.84
Max DD (%)	-18.5	-27.2

Interpretation. 1/N is often a surprisingly strong baseline, but RA-GA outperforms it heavily in Sharpe and significantly reduces drawdowns, proving that RA-GA’s regime-awareness adds considerable value beyond naive diversification.

RA-GA vs 60/40

Table 5: Ablation vs 60/40 benchmark.

Metric	RA-GA	60/40
Sharpe	1.23	0.78
Max DD (%)	-18.5	-29.4

Interpretation. 60/40 suffers during periods when both equities and bonds fall (2022). RA-GA avoids this correlation trap by rotating into gold and intermediate-duration treasuries.

RA-GA vs HRP

Table 6: Ablation vs HRP.

Metric	RA-GA	HRP
Sharpe	1.23	0.91
Max DD (%)	-18.5	-23.1

Interpretation. HRP diversifies well under stable conditions but reacts slowly to volatility spikes. RA-GA’s adaptivity leads to better protection and higher Sharpe.

RA-GA vs BTC-Only

Table 7: Ablation vs BTC-only strategy.

Metric	RA-GA	BTC
Sharpe	1.23	1.01
Max DD (%)	-18.5	-67.4

Interpretation. BTC-only portfolios are extremely volatile, with catastrophic drawdowns. RA-GA achieves BTC-like upside in many rallies but avoids downside collapses.

5.5 Bootstrap Robustness Analysis

To evaluate statistical robustness, a bootstrap procedure resamples annual return blocks with replacement to generate thousands of synthetic market histories. Table 8 summarizes the percentage of bootstrap iterations in which RA-GA outperforms each benchmark.

Table 8: Bootstrap outperformance probability across 10,000 resamples.

Comparison	Outperformance Probability
RA-GA vs 60/40	92%
RA-GA vs HRP	88%
RA-GA vs 1/N	97%

Interpretation.

1) RA-GA vs 60/40 (92%)

RA-GA almost always outperforms 60/40, even under random permutations of market conditions. This reveals robustness to macroeconomic uncertainty.

2) RA-GA vs HRP (88%)

The improvement over HRP demonstrates the importance of adaptive leverage and regime-awareness beyond simple diversification.

3) RA-GA vs 1/N (97%)

RA-GA decisively surpasses naive diversification, proving that its performance is not a fluke of the exact historical sequence.

4) Overall Conclusion

The bootstrap analysis provides strong statistical evidence that RA-GA's performance advantage is structural rather than path-dependent or due to luck. It reliably adapts even when market conditions are reshuffled in extreme ways.

5.6 Summary

Across NAV comparison, risk metrics, allocation dynamics, ablations, and bootstrap analysis, RA-GA consistently demonstrates:

- strong risk-adjusted performance,
- robust crisis behavior,
- smooth allocation transitions,
- generalization across synthetic resamples,
- and dominance over classical benchmarks.

These findings confirm that the RA-GA framework is a reliable solution for regime-aware global multi-asset portfolio construction.

6 Discussion

This section provides a deeper analysis of why the RA-GA framework performs well, its practical implementation advantages, its structural limitations, and failure modes of alternative approaches (including HMMs). We conclude with potential future extensions of the methodology.

6.1 Why RA-GA Works

The strong empirical performance of RA-GA arises from several interacting mechanisms that address well-known weaknesses in traditional optimizers.

1) Explicit Regime Awareness

Most portfolio optimizers implicitly assume stationarity. RA-GGA explicitly models market turbulence via realized volatility regimes, enabling the optimizer to:

- reduce exposure during turbulent markets,
- increase risk during calm markets,
- reallocate into defensive assets during drawdowns.

This adaptivity explains the significant drawdown compression observed in Section V.

2) Evolutionary Global Search

Classical MVO and HRP rely on deterministic optimization, which can trap the solution in local optima or overfit to historical noise. The evolutionary approach:

- explores a wide search space,
- escapes local minima,
- finds stable structures overlooked by convex methods,
- adapts naturally to non-linearities and non-Gaussian distributions.

These capabilities are essential in multi-asset markets where correlations and volatility regimes shift abruptly.

3) Turnover-Aware Fitness

By penalizing turnover directly, RA-GA produces allocations that:

- transition smoothly,
- avoid whipsaw trading,
- remain realistic for institutional investors.

This is a major advantage over MVO, which is known to generate unstable day-to-day allocations.

4) Leverage Scaling

Dynamic leverage enhances performance by:

- boosting returns during low-volatility periods,
- protecting the portfolio in crises via deleveraging.

This mechanism explains RA-GA's resilience during the COVID-19 crash and the 2022 bond–equity correlation breakdown.

5) Ensemble Averaging

Evolutionary algorithms can exhibit sample-path variance. Averaging two independent evolutionary runs:

- stabilizes allocations,
- reduces noise,
- improves generalization.

This contributes to the smooth NAV curve seen in Fig. [3](#).

6.2 Practical Implementation Advantages

RA-GA offers several advantages that make it suitable for real-world deployment in PMS, hedge funds, and multi-asset portfolios.

1. **Low turnover:** Typical monthly turnover remains around 6–9%, enabling cost-efficient trading.
2. **Interpretability:** Regime-aware shifts are economically intuitive—risk-off to bonds/precious metals, risk-on to equities.
3. **Scalability:** The method is compatible with larger asset universes and parallelizable evolutionary operations.
4. **Risk stability:** By adjusting leverage, RA-GA delivers a smooth volatility profile with minimal large drawdowns.
5. **FX compatibility:** Full INR normalization allows combined domestic and global portfolios without currency artifacts.
6. **Cross-regime robustness:** RA-GA behaves consistently across substantially different macro environments.

These characteristics make RA-GA appealing for investment managers seeking a systematic, low-volatility alternative to classical allocation strategies.

6.3 Limitations and Drawbacks

Despite its strong empirical performance, the RA-GA methodology is not without limitations. Understanding these drawbacks is important for realistic adoption and future extensions.

1) Computational Cost

Evolutionary optimization is computationally expensive relative to analytic methods such as MVO or HRP. With 80 individuals, 50 generations, and multiple ensemble runs, RA-GA requires several minutes per evaluation even on modern hardware.

2) Hyperparameter Sensitivity

Although GA is robust, hyperparameters such as mutation rate, tournament size, and crossover intensity affect convergence and solution diversity. Suboptimal choices can lead to premature convergence or overly random solutions.

3) Simplistic Regime Boundary

Median-split realized volatility is robust but simplistic. More granular regimes (e.g., three or four levels) may capture additional structure. However, overly complex regime models risk overfitting.

4) No Explicit Transaction Cost Model

Turnover penalties proxy trading costs, but RA-GA does not incorporate bid–ask spreads, market impact, or slippage explicitly.

5) Liquidity and Capacity Constraints

The method assumes infinite capacity and ignores liquidity limits important for large AUM deployments. Certain ETFs may experience slippage at high volume.

6) Ensemble Dilution

Averaging multiple GA runs stabilizes results but can dilute the best solutions. A weighted ensemble might improve performance.

7) Limited Use of Forward-Looking Signals

RA-GA uses only historical returns and realized volatility. It does not incorporate:

- macroeconomic indicators,
- earnings trends,
- yield curve dynamics,
- forward volatility expectations (VIX),
- macro trends such as inflation or unemployment.

This limits predictive power in fundamentally driven markets.

6.4 Failure Modes of Alternative Approaches (Including HMMs)

During this research, multiple alternative models were tested and failed to deliver satisfactory results. This section evaluates why.

1) Hidden Markov Models (HMMs)

HMMs were implemented for volatility regime detection, but they failed due to:

- **unstable state switching,**
- **non-convergence during high-volatility episodes,**
- **high sensitivity to initialization,**
- **economically meaningless regimes,**
- **failure to detect fast volatility shocks.**

In many cases, the HMM identified spurious regime changes or entered degenerate states with no clear separation between “normal” and “turbulent” conditions.

2) Reinforcement Learning

RL agents such as PPO and DQN were implemented. They failed due to:

- extreme sensitivity to hyperparameters,
- non-stationarity of market dynamics,
- unstable policy updates during crises,
- lack of generalization across regimes.

3) Classical Optimizers

Classical MVO and HRP failed in:

- highly correlated markets (2018Q4, 2022),
- sudden volatility spikes,
- environments with rapidly shifting covariance matrices.

These failures justify the development of a more adaptive method such as RA-GA.

6.5 Future Research Directions

Several extensions could improve the robustness and predictive power of RA-GA.

1) ML-Based Regime Detection

Instead of median-split realized volatility, one could use:

- transformer-based volatility forecasters,
- LSTM models for realized volatility,
- gradient-boosted trees (XGBoost) using macro features,
- neural volatility filters.

Such models may improve regime identification during complex macro environments.

2) Multiobjective Evolutionary Optimization

Future work could incorporate multiobjective GA frameworks (e.g., NSGA-II), optimizing:

- Sharpe ratio,
- maximum drawdown,
- volatility,
- turnover,
- tail-risk metrics (CVaR).

3) Bayesian GA

Introducing uncertainty-aware priors into fitness evaluation may stabilize allocations when datasets are short or when structural breaks occur.

4) Liquidity- and Cost-Aware Constraints

RA-GA could explicitly incorporate:

- transaction costs,
- market impact,
- liquidity filters,
- trade-size constraints.

This would be essential for large AUM deployment.

5) Adaptive Mutation and Crossover

Mutation and crossover parameters could evolve dynamically based on:

- population diversity,
- convergence rates,
- recent market volatility.

This would allow RA-GA to adjust exploration intensity on the fly.

6) Online Learning and Rolling GA Updates

Instead of monthly static reoptimization, an online GA could update:

- regime thresholds,
- leverage levels,
- weight distributions,
- mutation intensities

continuously as new data arrives.

7) Macro-Aware Fitness

Integrating macroeconomic indicators (inflation, yields, credit spreads) into the fitness function could improve predictive allocation decisions.

6.6 Summary

RA-GA achieves strong performance by combining regime awareness, evolutionary global search, turnover control, and dynamic leverage scaling. While the framework has limitations and can be extended further, its present formulation provides a robust foundation for regime-aware multi-asset allocation.

7 Conclusion

This paper introduced a Regime-Aware Genetic Algorithm (RA-GA) for global multi-asset portfolio optimization. The methodology integrates three essential components: realized-volatility regime detection, dynamic leverage scaling, and evolutionary search with turnover constraints and ensemble averaging. Unlike classical optimizers—such as MVO, Black–Litterman, risk-parity, and HRP—which often assume stationary market conditions, RA-GA explicitly accounts for time-varying market risk and dynamically adapts to structural breaks. The framework is computationally intensive, but its flexibility and robustness make it well-suited for real-world scenarios where non-linearity, noise, and regime-dependence dominate asset behavior.

The empirical results from 2016–2025 show that RA-GA delivers superior risk-adjusted performance compared with 60/40, HRP, 1/N, and BTC-only baselines. It achieves a higher Sharpe ratio, lower volatility, and substantially reduced drawdowns, with particularly strong resilience during crises such as the 2018Q4 selloff, the 2020 COVID-19 crash, and the 2022 global inflation shock. Its dynamic rotation between equities, bonds, and commodities reveals a behaviorally intuitive pattern that aligns with common macroeconomic cycles: equity dominance in calm markets and defensive positioning in turbulent ones.

In addition to performance, RA-GA exhibits practical implementability through low turnover, stable allocation transitions, and full INR normalization for cross-currency portfolios. Bootstrap experiments demonstrate that RA-GA outperforms benchmark strategies in a large majority of synthetic market paths, indicating that its performance advantage is structural rather than historically path-dependent.

Despite its strengths, RA-GA has limitations. It lacks explicit modeling of transaction costs, liquidity constraints, and macroeconomic indicators, and its realized-volatility regime classifier is intentionally simple. Furthermore, evolutionary optimization inevitably incurs computational overhead and can be sensitive to hyperparameters. Nevertheless, these drawbacks offer fertile ground for future research. Promising extensions include machine-learning-driven regime detection (e.g., LSTM-based volatility models or transformer forecasters), multiobjective evolutionary optimization (e.g., NSGA-II), liquidity-aware constraints, Bayesian evolutionary search, and online adaptive GA frameworks.

Overall, RA-GA demonstrates that combining evolutionary search with explicit regime-awareness provides a powerful and practical approach to multi-asset allocation. It offers a compelling alternative to both classical and modern data-driven strategies, and it lays a scalable foundation for future research in robust, adaptive portfolio construction.

Appendix

A Genetic Algorithm Pseudocode

The following pseudocode summarizes the monthly RA-GA optimization procedure:

[h] [1] **Input:** Price data P_t , returns r_t , covariance Σ , previous weights w_{t-1} Compute realized volatility σ_t using 21-day lookback Determine regime: $\sigma_t \leq \text{median}(\sigma) \Rightarrow$ low-vol; otherwise high-vol Select leverage L_t based on regime Initialize population of portfolios $\{w^{(i)}\}_{i=1}^{\text{POP}}$ generation = 1 to G Evaluate fitness $F(w^{(i)})$ for each individual:

$$F(w) = \frac{\mathbb{E}[w^\top r_t]}{\sqrt{w^\top \Sigma w}} - \alpha \cdot \text{Turnover}(w)$$

Select parents via tournament selection ($k = 3$) Apply BLX- α crossover to produce offspring Apply Gaussian mutation to random genes Enforce weight constraints and normalization Carry top 2 elite individuals forward Let w^* be the best solution Apply leverage scaling: $w_t^{\text{scaled}} = L_t \cdot w^*$ **Output:** Final monthly allocation w_t

B NAV Computation

Monthly NAV is computed by compounding returns:

$$\text{NAV}_t = \text{NAV}_{t-1} \left(1 + \sum_{i=1}^N w_{i,t} \cdot r_{i,t} \right),$$

where $w_{i,t}$ is the allocation to asset i at time t , $r_{i,t}$ is the return of asset i at time t , and $\text{NAV}_0 = 1$.

C Turnover Computation

Turnover captures the trading volume required to move from one month's allocation to the next:

$$\text{Turnover}(t) = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|.$$

This term is included in the fitness function to penalize excessive trading.

D Benchmark Definitions

60/40

60% equities (SPY) + 40% intermediate Treasuries (IEF), monthly rebalanced.

HRP

Hierarchical Risk Parity constructed per [4] on the 26-asset universe, monthly rebalanced.

1/N

Equal-weight across all 26 assets, monthly rebalanced.

BTC-only

Fully invested in Bitcoin.

E Bootstrap Procedure

We run a block bootstrap on annual return blocks:

1. Extract annual return vectors for each year in the sample.
2. For each of 10,000 iterations:
 - (a) Sample years with replacement to create a synthetic sequence.
 - (b) Recompute NAVs for RA-GA and benchmarks along the synthetic path.
 - (c) Record performance differentials (RA-GA minus baseline).
3. Aggregate the differential distribution and compute empirical outperformance probabilities and confidence intervals.

This preserves within-year temporal dependence while testing robustness to different year-orderings.

F Hyperparameters Used in RA-GA

Table 9: RA-GA hyperparameters (used in experiments).

Parameter	Value
Population size	80
Generations	50
Tournament size (k)	3
Mutation rate	0.20
Mutation std	0.03
Elitism count	2
Ensemble runs	2
Low-vol leverage	1.2–1.4
High-vol leverage	0.6–0.8
Turnover penalty (α)	0.10

References

- [1] H. Markowitz, “Portfolio selection,” *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
- [2] O. Ledoit and M. Wolf, “Improved estimation of the covariance matrix of stock returns with an application to portfolio selection,” *Journal of Empirical Finance*, vol. 10, no. 5, pp. 603–621, 2003.
- [3] S. Maillard, T. Roncalli, and J. Teïletche, “The properties of equally weighted risk contributions portfolios,” *The Journal of Portfolio Management*, vol. 36, no. 4, pp. 60–70, 2010.
- [4] M. López de Prado, “Building diversified portfolios that outperform out of sample,” *The Journal of Portfolio Management*, vol. 42, no. 4, pp. 59–69, 2016.
- [5] J. D. Hamilton, “Analysis of time series subject to changes in regime,” *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.

- [6] A. Ang and G. Bekaert, “International asset allocation with regime shifts,” *The Review of Financial Studies*, vol. 15, no. 4, pp. 1137–1187, 2002.
- [7] J. H. Holland, *Adaptation in Natural and Artificial Systems*. University of Michigan Press, 1975.
- [8] T.-J. Chang, N. Meade, J. E. Beasley, and Y. M. Sharaiha, “Heuristics for cardinality constrained portfolio optimisation,” *Computers & Operations Research*, vol. 27, no. 13, pp. 1271–1302, 2000.
- [9] Y. Crama and H. L. Schyns, “Simulated annealing for complex portfolio selection problems,” *European Journal of Operational Research*, vol. 150, no. 3, pp. 546–571, 2003.